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Model used: gemini-3.5-flash-lite

01 Target Distribution

What it shows: The bar chart displays the frequency count of the target variable 'Churn' on the y-axis (Number of Rows) against two binary categories on the x-axis: Negative (0) and Positive (1).

Key finding: The dataset contains 33,881 non-churned customers (52.6%) and 30,493 churned customers (47.4%), indicating a nearly balanced class distribution.

Business implication: Because the classes are well-balanced, predictive models will not require heavy resampling techniques like SMOTE to prevent severe class imbalance bias.

Confidence caveat: This distribution reflects raw row counts, which may include duplicate records or oversampled data rather than a natural active customer base.

Suggestion: Verify if this dataset has been artificially balanced or augmented; if it represents a raw operational sample, investigate why the churn rate is exceptionally high at 47.4%.

02 Rate By Gender

What it shows: The horizontal bar chart displays the positive rate percentage split by customer gender (Female and Male). The x-axis represents the positive rate percentage ranging from 0 to 50+, while the y-axis shows the gender categories.

Key finding: Female customers exhibit a significantly higher positive rate of 55.0%, compared to Male customers at 38.6%.

Business implication: Retention teams should investigate the underlying drivers for this demographic disparity to prevent potential churn risks concentrated among female subscribers.

Confidence caveat: Gender alone does not imply direct causation of churn; multivariate analysis controlling for contract type and monthly charges is required to confirm if gender is an independent risk factor.

Suggestion: Do not allocate retention budgets based solely on gender demographics; instead, cross-reference gender with high-risk variables like payment method and monthly charges to target actionable churn drivers.

03 Rate By Subscription Type

What it shows: This horizontal bar chart displays the positive rate percentage on the x-axis

across three distinct subscription types (Basic, Standard, and Premium) on the y-axis.

Key finding: The positive rate shows minimal variation across tiers, with Basic at 48.3%, Standard at 47.3%, and Premium at 46.5%.

Business implication: Retention and marketing teams should not rely on subscription tier adjustments alone to drive churn reduction, as plan type has virtually no impact on churn rates.

Confidence caveat: This chart shows correlation rather than causation, and sample sizes per tier should be verified before restructuring pricing models.

Suggestion: Analyze churn rates by combining subscription types with contract lengths and monthly charges to identify higher-risk customer segments.

04 Rate By Contract Length

What it shows: This horizontal bar chart displays the positive rate percentage on the x-axis against three contract length categories (Monthly, Annual, and Quarterly) on the y-axis. Each horizontal bar represents the specific positive rate value for that contract type.

Key finding: Monthly contracts exhibit the highest positive rate at 51.6%, whereas Quarterly contracts show the lowest positive rate at 44.0%, with Annual contracts falling in between at 46.2%.

Business implication: The retention team should prioritize targeted engagement campaigns for customers on monthly contracts, as they demonstrate significantly higher vulnerability. Incentivizing these users to transition to longer-term commitments can effectively reduce overall churn risk.

Confidence caveat: This analysis relies purely on contract length proportions without controlling for confounding variables like monthly charges or customer tenure.

Suggestion: Cross-reference contract lengths with monthly charges and tenure segments to isolate whether contract type is the primary driver or if pricing sensitivity heavily influences these rates.

05 Distribution Age

What it shows: The chart displays overlaid histograms representing the probability density distribution of customer age, segmented by churn status ("Negative" in blue, "Positive" in orange). Vertical dashed lines indicate the respective median ages for each group.

Key finding: The median age for churned customers (Positive) is 44.0 years, compared to 41.0 years for retained customers (Negative). The chart reveals a slightly older demographic profile among churned users, particularly visible in bins above age 50.

Business implication: Retention teams should investigate potential age-specific pain points, such as pricing sensitivity or digital service usability for older customers. Tailored loyalty programs or specialized support channels could mitigate churn in the 50+ age bracket.

Confidence caveat: Age differences are relatively minor and may overlap with confounding

variables like contract type or tenure rather than serving as an independent driver of churn.

Suggestion: Cross-tabulate age groups with contract type and internet service tiers to determine if older customers predominantly select specific high-churn plans (e.g., month-to-month fiber optic).

06 Boxplot Age

What it shows: The chart displays two boxplots comparing customer age (vertical axis) against the target group (Negative vs. Positive churn status on the horizontal axis). It visualizes the distribution, median, and interquartile range of ages for both cohorts.

Key finding: The median age for the Positive churn group is approximately 44 years, compared to roughly 41 years for the Negative group. Overall age distributions between churned and retained customers show substantial overlap from 18 to 65 years.

Business implication: Age is not a strong primary differentiator for customer churn, meaning the retention team should not allocate budget toward age-targeted retention campaigns. Instead, focus retention efforts on behavioral metrics like contract type and monthly charges.

Confidence caveat: This bivariate analysis does not control for confounding variables such as tenure or pricing plans, which may mask underlying age-related churn segments.

Suggestion: Combine age with tenure and contract type in a multivariate logistic regression model to test for non-linear or segmented age interactions on churn.

07 Distribution Tenure

What it shows: The chart displays a stacked density histogram comparing customer tenure (in months) between churned (Negative/blue) and retained (Positive/orange) customers, with vertical dashed lines marking their respective medians.

Key finding: Churned customers have a significantly lower median tenure of 27.0 months compared to 37.0 months for retained customers, with churn heavily concentrated in the first 20-25 months.

Business implication: The retention team must focus early intervention efforts on the critical onboarding window within the first two years when churn risk peaks.

Confidence caveat: This visualization shows correlation only; tenure may be confounded by contract type or pricing plans that require multivariate analysis to isolate.

Suggestion: Implement proactive check-ins, targeted loyalty incentives, and contract-length optimization campaigns during months 1 through 24 to dramatically lower early-stage churn.

08 Boxplot Tenure

What it shows: The chart displays side-by-side boxplots comparing customer tenure in months across two target groups: Negative (non-churned) and Positive (churned). The y-axis

represents tenure ranging from 0 to 60+ months.

Key finding: Non-churned customers (Negative) have a significantly lower median tenure of approximately 27 months compared to churned customers (Positive), who show a median tenure of around 37 months. The interquartile range for churned customers spans from roughly 25 to 49 months.

Business implication: Retention teams should focus early engagement efforts on newer cohorts, but also investigate why customers churning at higher tenures are leaving after establishing multi-year relationships.

Confidence caveat: This chart shows correlation rather than causation, and additional variables like monthly charges or contract type should be cross-analyzed to understand the true drivers behind mid-to-long-term churn.

Suggestion: Implement targeted loyalty milestones and satisfaction checks between the 24-month and 48-month marks to mitigate the unexpected loss of higher-tenure customers.

09 Correlation Heatmap

What it shows: This chart is a correlation matrix heatmap displaying pairwise Pearson correlation coefficients between numerical customer attributes and the churn target flag, with values ranging from -1.0 to 1.0 encoded via a red-to-blue diverging color scale.

Key finding: `payment_delay` shows the strongest positive correlation with churn (0.56), followed by `support_calls` (0.30) and `tenure` (0.20), while `usage_frequency` shows a negative correlation (-0.12).

Business implication: The retention team should implement early-warning triggers for customers exhibiting payment delays and frequent support calls to intervene before they churn.

Confidence caveat: Correlation does not imply causation; these linear relationships require multivariate modeling to control for confounding variables before allocating intervention budgets.

Suggestion: Build a predictive classification model (e.g., Logistic Regression or XGBoost) using these features to calculate individual customer churn probabilities and automate targeted retention offers.

10 Combo Gender Subscription Type

What it shows: The clustered bar chart displays the positive rate percentage on the y-axis, grouped by gender (Female vs. Male) on the x-axis, with each cluster broken down by subscription type (Basic, Premium, Standard).

Key finding: Female customers exhibit significantly higher positive rates across all subscription types, hovering around 54% to 56%, whereas male customers consistently lag behind at approximately 38% to 40%.

Business implication: The retention team should investigate why male customers across all tiers have lower positive engagement and develop targeted retention campaigns tailored

specifically to this segment.

Confidence caveat: This chart shows correlation by demographic segment; adding tenure and pricing data is necessary to determine if gender drives the variance or if it correlates with underlying plan usage.

Suggestion: Conduct a multivariate analysis combining gender with contract type and monthly charges to uncover the root causes of the demographic performance gap.